Recitation 3

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Outline

1. **Homework**
2. **Review**
3. **Example**
4. **Recitation**

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1. **Homework**

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4. Assume that each child is, independently, equally likely to be either a boy or a girl. If every family has three children, what proportion of all sons are eldest sons?

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Solution:

Letting E be the event that the child is an eldest son, letting S be the event that the child is a son. The answer for the problem is

$$\mathbf{P}(E \mid S) = \frac{\mathbf{P}(E \cap S)}{\mathbf{P}(S)} = 2\mathbf{P}(E)$$

with $\mathbf{P}(E \cap S) = \mathbf{P}(E)$ and $\mathbf{P}(S) = 1/2$.
Letting A be the event that the child's family has at least one son. With total probability theorem,

$$\mathbf{P}(E \mid S) = 2\left[\mathbf{P}(E \mid A)\mathbf{P}(A) + \mathbf{P}(E \mid A^c)\mathbf{P}(A^c)\right] = \frac{7}{12}$$

with $\mathbf{P}(A^c) = \mathbf{P}(\text{three daughters}) = 1/8$ and $\mathbf{P}(E \mid A) = 1/3$.

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- Independence and conditional independence of two events

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$
$$\mathbf{P}(A \cap B \mid C) = \mathbf{P}(A \mid C) \cdot \mathbf{P}(B \mid C)$$

- Independence of a collection of events

$$\mathbf{P}(A_i \cap A_j \cap \cdots \cap A_q) = \mathbf{P}(A_i)\mathbf{P}(A_j) \cdots \mathbf{P}(A_q)$$

for **any** distinct indices $i, j, \ldots, q$ chosen from $\{1, \ldots, n\}$

- Counting results:
  - Permutations of n objects: $n!$
  - k-permutations of n objects: $n!/(n-k)!$
  - Combinations of k out of n objects:

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

  - Partitions of n objects into r groups, with the ith group having n_i objects:

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1!\, n_2! \cdots n_r!}$$

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Combinations

- $\binom{n}{k}$: number of k-element subsets of a given n-element set
- Two ways of constructing an ordered sequence of k **distinct** items:
  - Choose the k items one at a time:

$$n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!} \text{ choices}$$

  - Choose k items, then order them ($k!$ possible orders)
- Hence:

$$\binom{n}{k} \cdot k! = \frac{n!}{(n-k)!}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Coin tossing problem

- Event B: 3 out of 10 tosses were "heads".
  - Given that B occurred, what is the (conditional) probability that the first 2 tosses were heads?
- All outcomes in set B are equally likely: probability $p^3(1-p)^7$
  - Conditional probability law is uniform
- Number of outcomes in B:

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- Out of the outcomes in B, how many start with HH?
  - $\binom{10-2}{3-2} = \binom{8}{1}$

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- Count number of ways of dealing the remaining 48 cards

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- Count number of ways of distributing the four aces: $4 \cdot 3 \cdot 2$
- Count number of ways of dealing the remaining 48 cards

$$\frac{48!}{12!\,12!\,12!\,12!}$$

- Answer:

$$\frac{4 \cdot 3 \cdot 2 \cdot \dfrac{48!}{12!\,12!\,12!\,12!}}{\dfrac{52!}{13!\,13!\,13!\,13!}}$$

Counting Results

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- Partitions of n objects into r groups, with the ith group having n_i objects:

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4. **Recitation**

1. About events.

(a) Can an event A be independent of itself?

(b) Let A and B be independent events. Use the definition of independence to prove that the events A and B^c are independent.

(c) Let A, B, and C be independent events, with $\mathbf{P}(C) > 0$. Prove that A and B are conditionally independent of C.

1. About events.

(a) Can an event A be independent of itself?

A is independent of itself if and only if $\mathbf{P}(A \cap A) = \mathbf{P}(A)\mathbf{P}(A)$. Since $A \cap A = A$ then A must satisfy $\mathbf{P}(A) = (\mathbf{P}(A))^2$. Therefore, A is independent of itself if and only if $\mathbf{P}(A) = 1$ or $\mathbf{P}(A) = 0$.

1. About events.

(b) Let A and B be independent events. Use the definition of independence to prove that the events A and B^c are independent.

Using the additivity axiom and the independence of A and B, we obtain

$$\mathbf{P}(A) = \mathbf{P}(A \cap B) + \mathbf{P}(A \cap B^c) = \mathbf{P}(A)\mathbf{P}(B) + \mathbf{P}(A \cap B^c).$$

It follows that

$$\mathbf{P}(A \cap B^c) = \mathbf{P}(A)(1 - \mathbf{P}(B)) = \mathbf{P}(A)\mathbf{P}(B^c).$$

1. About events.

(c) Let A, B, and C be independent events, with $\mathbf{P}(C) > 0$. Prove that A and B are conditionally independent of C.

We have

$$\begin{aligned}\mathbf{P}(A \cap B|C) &= \frac{\mathbf{P}(A \cap B \cap C)}{\mathbf{P}(C)} \\ &= \frac{\mathbf{P}(A)\mathbf{P}(B)\mathbf{P}(C)}{\mathbf{P}(C)} \\ &= \mathbf{P}(A)\mathbf{P}(B) \\ &= \mathbf{P}(A|C)\mathbf{P}(B|C).\end{aligned}$$

2. Consider two independent fair coin tosses, in which all four possible outcomes are equally likely. Let

$$H_1 = \{\text{1st toss is a head}\},$$
$$H_2 = \{\text{2nd toss is a head}\},$$
$$D = \{\text{the two tosses produced different results}\}.$$

(*a*) Are the events H_1 and H_2 (unconditionally) independent?

(*b*) Given event D has occurred, are the events H_1 and H_2 (conditionally) independent?

2. Consider two independent fair coin tosses, in which all four possible outcomes are equally likely. Let

$$
\begin{aligned}
H_1 &= \{\text{1st toss is a head}\},\\
H_2 &= \{\text{2nd toss is a head}\},\\
D &= \{\text{the two tosses produced different results}\}.
\end{aligned}
$$

The events H_1 and H_2 are (unconditionally) independent. But

$$\mathbf{P}(H_1|D) = \frac{1}{2}, \quad \mathbf{P}(H_2|D) = \frac{1}{2}, \quad \mathbf{P}(H_1 \cap H_2|D) = 0,$$

so that $\mathbf{P}(H_1 \cap H_2|D) \neq \mathbf{P}(H_1|D)\mathbf{P}(H_2|D)$, and H_1, H_2 are not conditionally independent.

3. Imagine a drunk tightrope walker, in the middle of a really long tightrope, who manages to keep his balance, but takes a step forward with probability p and takes a step back with probability $(1-p)$.

(a) What is the probability that after two steps the tightrope walker will be at the same place on the rope?

(b) What is the probability that after three steps, the tightrope walker will be one step forward from where he began?

(c) Given that after three steps he has managed to move ahead one step, what is the probability that the first step he took was a step forward?

(a) What is the probability that after two steps the tightrope walker will be at the same place on the rope?
In order to wind up in the same place after two steps, the tightrope walker can either step forwards, then backwards, or vice versa. Therefore the required probability is:

$$2 \cdot p \cdot (1-p).$$

(b) What is the probability that after three steps, the tightrope walker will be one step forward from where he began?
The probability that after three steps he will be one step ahead of his starting point is the probability that out of 3 steps in total, 2 of them are forwards, and one is backwards. This equals:

$$3 \cdot p^2 \cdot (1-p).$$

(c) Given that after three steps he has managed to move ahead one step, what is the probability that the first step he took was a step forward?

Given that out of his three steps only one is backwards, the sample space for the experiment is:

$$\{(F, F, B); (F, B, F); (B, F, F)\}$$

where F denotes a step forwards, and B a step backwards. Each of these sample points is equally likely, therefore the probability that his first step is a step forward is $\frac{2}{3}$.